

13_January_2023 • Unit 2

1H + 2H source-order past paper

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L-to-M | CONVERTED MODULAR EDITION

Selected from Linear Higher papers; not an original Modular examination paper.

Contents

- 1H Q01 • Statistics Toolkit • 4 marks
- 1H Q03 • Bearings, Scale Drawing & Constructions • 2 marks
- 1H Q04 • Angles in Polygons & Parallel Lines • 3 marks
- 1H Q05 • Ratio Problem Solving • 3 marks
- 1H Q06 • Percentages • 7 marks
- 1H Q09 • Powers, Roots & Standard Form • 3 marks
- 1H Q10 • Solving Inequalities • 6 marks
- 1H Q11 • Graphing Inequalities • 3 marks
- 1H Q17 • Direct & Inverse Proportion • 3 marks
- 1H Q18 • Functions • 4 marks
- 1H Q20 • Differentiation • 5 marks
- 1H Q22 • Sequences • 4 marks
- 1H Q24 • Vectors • 5 marks
- 2H Q07 • Simultaneous Equations • 3 marks
- 2H Q08 • Compound Interest & Depreciation • 4 marks
- 2H Q12 • Cumulative Frequency Diagrams • 5 marks
- 2H Q14 • Circle Theorems • 5 marks
- 2H Q15 • Statistics Toolkit • 3 marks
- 2H Q17 • Algebraic Proof • 4 marks
- 2H Q19 • Area & Volume of Similar Shapes • 4 marks
- 2H Q20 • Simultaneous Equations • 5 marks
- 2H Q21 • Volume & Surface Area • 3 marks
- 2H Q22 • Solving Inequalities • 3 marks
- 2H Q24 • Graphs of Functions • 2 marks
- 2H Q25 • Functions • 4 marks

Answer ALL TWENTY FOUR questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1** 80 students entered a dancing competition.

The table gives information about the length of time, in minutes, for which each student spent dancing.

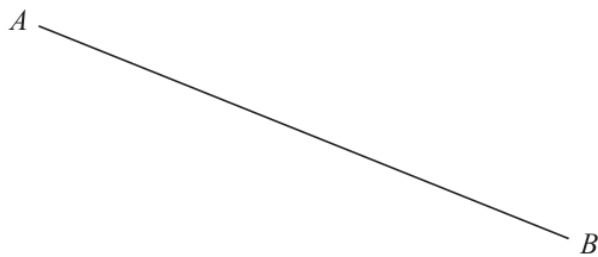
Time (m)	Frequency
$0 < m \leq 12$	11
$12 < m \leq 24$	25
$24 < m \leq 36$	23
$36 < m \leq 48$	15
$48 < m \leq 60$	6

Work out an estimate for the mean length of time the students spent dancing.

..... minutes

(Total for Question 1 is 4 marks)

- 3 Use ruler and compasses only to construct the perpendicular bisector of line AB
You must show all your construction lines.



(Total for Question 3 is 2 marks)

- 4 The diagram shows a pentagon.

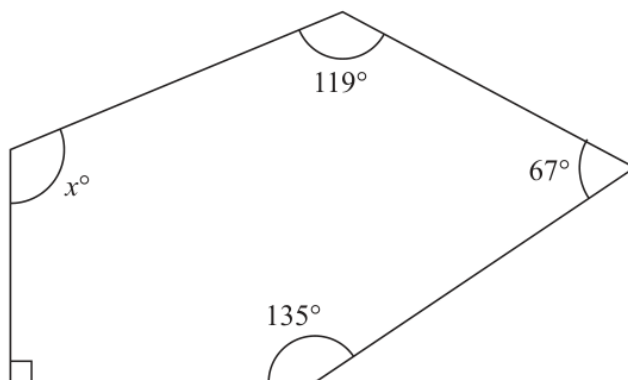


Diagram **NOT**
accurately drawn

Work out the value of x

$x =$

(Total for Question 4 is 3 marks)

- 5 In a box, there are only green sweets, orange sweets and yellow sweets.

There are 280 sweets in the box so that

the number of green sweets : the number of orange sweets = 2 : 3

and

the number of orange sweets : the number of yellow sweets = 1 : 5

Work out how many green sweets there are in the box.

(Total for Question 5 is 3 marks)

6 Shane bought a car.

The amount Shane paid for the car was \$32 000

Theresa also bought a car.

To pay for this car, Theresa paid a deposit of \$18 000 together with 14 monthly payments of \$1160

Theresa paid more for her car than Shane paid for his car.

(a) Work out how much more Theresa paid as a percentage of the amount Shane paid.

..... %
(4)

Kylie bought a van.

After 1 year, the value of the van was \$39 865

During this year, the value of the van decreased by 15%

(b) Work out the value of the van when Kylie bought it.

\$.....
(3)

(Total for Question 6 is 7 marks)

- 9 (a) Write 6.25×10^{-4} as an ordinary number.

.....
(1)

- (b) Work out $(2.4 \times 10^{12}) \div (9.6 \times 10^4)$
Give your answer in standard form.

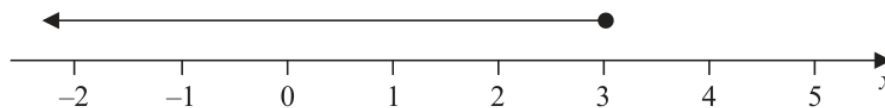
.....
(2)

(Total for Question 9 is 3 marks)

10 (a) Factorise $y^2 - 2y - 48$

.....
(2)

(b) Write down the inequality shown on the number line



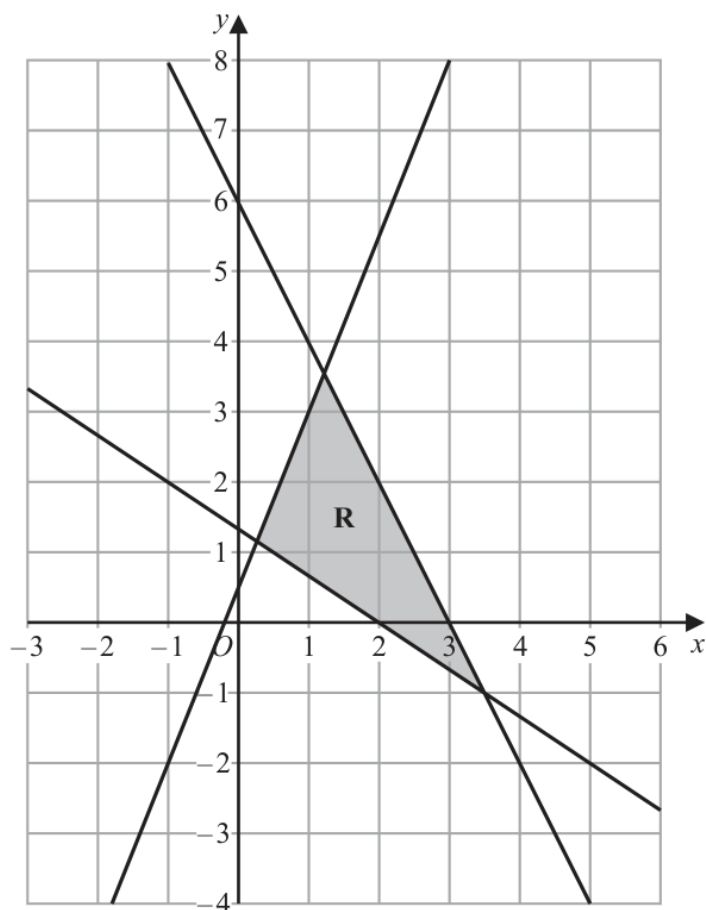
.....
(1)

(c) Solve the inequality $7w + 6 > 12w + 14$

.....
(3)

(Total for Question 10 is 6 marks)

11



The region **R**, shown shaded in the diagram, is bounded by the straight lines with equations

$$2x + y = 6$$

$$2y = 5x + 1$$

$$3y + 2x = 4$$

Write down the three inequalities that define **R**

.....

.....

.....

(Total for Question 11 is 3 marks)

17 y is inversely proportional to $\sqrt{x}$

$y = c^4$ when $x = c^2$ where c is a positive constant.

Find a formula for y in terms of x and c
Give your answer in its simplest form.

(Total for Question 17 is 3 marks)

18 The function f is such that $f(x) = \frac{k}{x}$ where $x \neq 0$ and k is an integer.

(a) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$$f^{-1}(x) = \dots \quad (1)$$

The function g is such that $g(x) = 2 - 3x^4$ where $x \neq 0$

The function h is such that $h(x) = \frac{3x}{2-x}$ where $x \neq 2$

(b) (i) Find $g(-2)$

$$\dots \quad (1)$$

(ii) Express the composite function hg in the form $hg(x) = \dots$
Give your answer in its simplest form.

$$hg(x) = \dots \quad (2)$$

(Total for Question 18 is 4 marks)

20 The radius of a right circular cylinder is x cm.

The height of the cylinder is $\left(\frac{800}{\pi x} - x\right)$ cm.

The volume of the cylinder is $V \text{ cm}^3$

Find the maximum value of V

Give your answer correct to the nearest whole number.



(Total for Question 20 is 5 marks)

- 22 The first term of an arithmetic series is $(2t + 1)$ where $t > 0$
The n th term of this arithmetic series is $(14t - 5)$

The common difference of the series is 3

The sum of the first n terms of the series can be written as $p(qt - 1)^r$ where p , q and r are integers.

Find the value of p , the value of q and the value of r
Show clear algebraic working.



$$p = \dots\dots\dots q = \dots\dots\dots r = \dots\dots\dots$$

(Total for Question 22 is 4 marks)

24 $OAED$ is a quadrilateral.

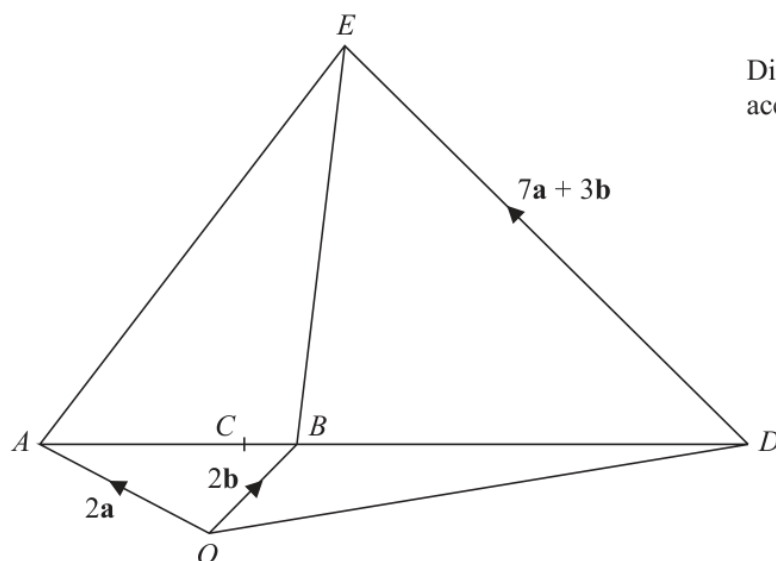


Diagram **NOT**
accurately drawn

$$\vec{OA} = 2\mathbf{a} \quad \vec{OB} = 2\mathbf{b} \quad \vec{DE} = 7\mathbf{a} + 3\mathbf{b}$$

$$AB:BD = 1:2$$

The point C on AD is such that OCE is a straight line.

Use a vector method to find the ratio of $OC:CE$



(Total for Question 24 is 5 marks)

7 Solve the simultaneous equations

$$\begin{aligned}5x + 4y &= -2 \\ 2x - y &= 4.4\end{aligned}$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 7 is 3 marks)

- 8 Matteo is going to invest 5000 Swiss francs for two years.

He can invest his money in Bank **G** or in Bank **H**.

Bank **G**
1.6% per year
compound interest

Bank **H**
2.9% interest added after
two years

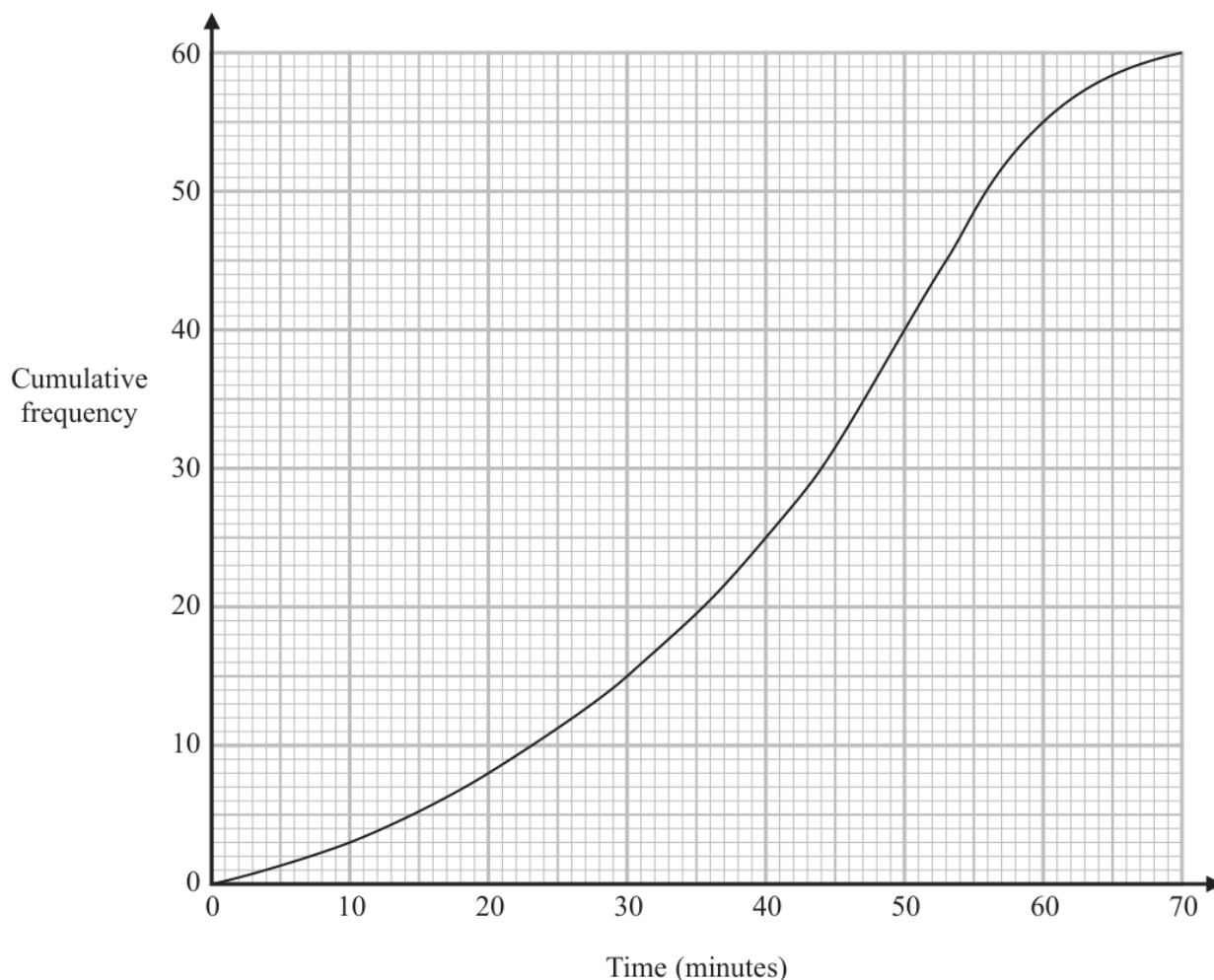
The total amount of interest Matteo would receive at the end of two years from Bank **G** is more than the amount of interest Matteo would receive at the end of two years from Bank **H**.

How much more?

..... Swiss francs

(Total for Question 8 is 4 marks)

- 12 The cumulative frequency graph gives information about the time, in minutes, each of 60 people took to shop in a market.



- (a) Use the graph to find an estimate for the median time people took to shop in the market.

..... minutes
(1)

- (b) Use the graph to find an estimate for the number of people who took longer than 55 minutes to shop in the market.

.....
(2)



- (c) Use the graph to complete the frequency table to give information about the time, in minutes, each of the 60 people took to shop in the market.

Time taken to shop in the market (m minutes)	Frequency
$0 < m \leq 10$	3
$10 < m \leq 20$	5
$20 < m \leq 30$	
$30 < m \leq 40$	
$40 < m \leq 50$	
$50 < m \leq 60$	
$60 < m \leq 70$	

(2)

(Total for Question 12 is 5 marks)

14 A, B, C and D are points on a circle, centre O

EBF is the tangent to the circle at B

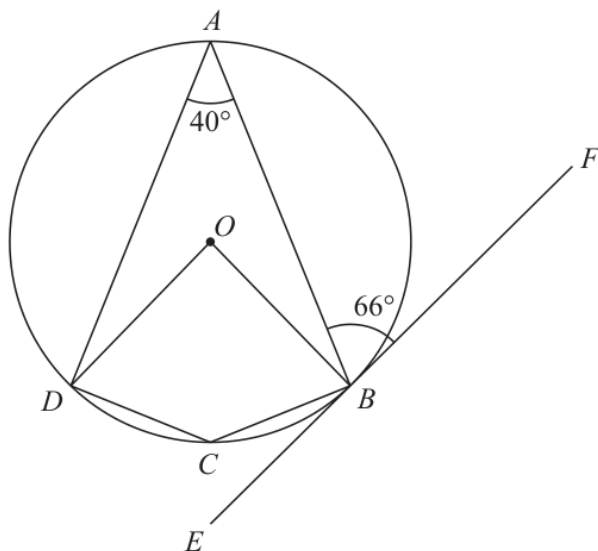


Diagram **NOT**
accurately drawn

(a) (i) Work out the size of angle DCB

o

(1)

(ii) Give a reason for your answer to (a)(i)

(1)

(b) Work out the size of angle ADO

o

(3)

(Total for Question 14 is 5 marks)

- 15** Here is a list giving the numbers of runs scored last week by the eleven members of cricket team **A**.

2 3 4 6 21 26 27 32 34 61 72

The interquartile range of the numbers of runs scored last week by the eleven members of cricket team **B** was 42

Using a suitable calculation, write down one comparison between the numbers of runs scored by the members of cricket team **A** and the members of cricket team **B**.
Show your working clearly.

(Total for Question 15 is 3 marks)

- 17 Given that $8\sqrt{m} + \sqrt{49m} - \sqrt{9m} = k\sqrt{m}$
where k is an integer and m is a prime number,

(a) work out the value of k

$$k = \dots\dots\dots (1)$$

- (b) Show that $\frac{5 - \sqrt{18}}{1 - \sqrt{2}}$ can be written in the form $a + b\sqrt{2}$

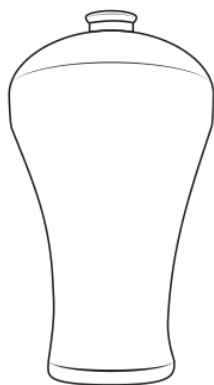
where a and b are integers.

Show each stage of your working clearly.

(3)

(Total for Question 17 is 4 marks)

19 **A** and **B** are two similar vases.



A



B

Diagram **NOT**
accurately drawn

The vases are such that

$$\text{surface area of vase } \mathbf{B} = \frac{25}{64} \times \text{surface area of vase } \mathbf{A}$$

and that

$$\text{volume of vase } \mathbf{A} - \text{volume of vase } \mathbf{B} = 541.8 \text{ cm}^3$$

Calculate the volume of vase **B**

..... cm^3

(Total for Question 19 is 4 marks)

20 Solve the simultaneous equations

$$\begin{aligned}y &= 7 - 2x \\ x^2 + y^2 &= 34\end{aligned}$$

Show clear algebraic working.

(Total for Question 20 is 5 marks)

- 21** Given that the surface area of a sphere is $49\pi\text{cm}^2$
find the volume of the sphere.
Give your answer correct to the nearest integer.

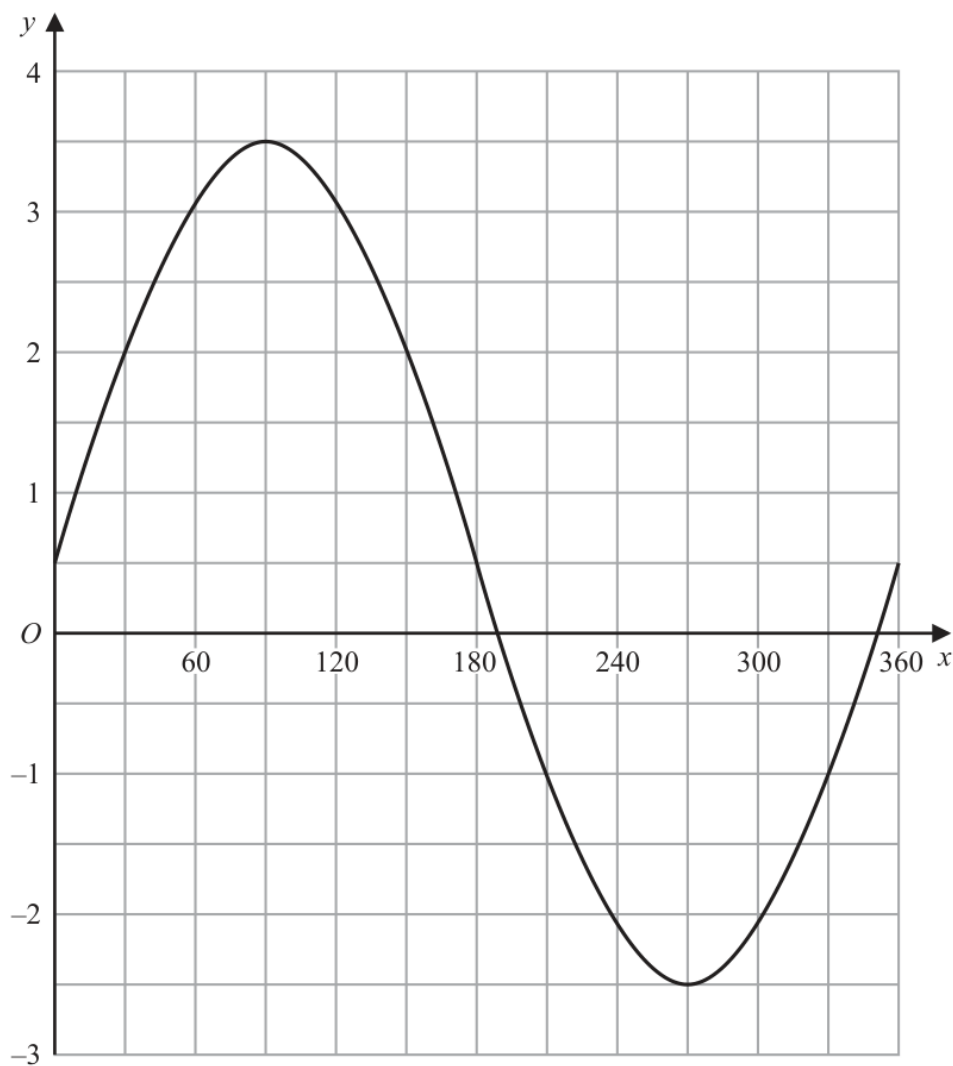
..... cm^3

(Total for Question 21 is 3 marks)

- 22** Solve the inequality $6x^2 + 37x \leq 35$
Show clear algebraic working.

(Total for Question 22 is 3 marks)

24 The graph of $y = a \sin x^\circ + b$ is drawn on the grid.



Find the value of a and the value of b

$a =$

$b =$

(Total for Question 24 is 2 marks)

25 The function f is such that $f(x) = 3x^2 - 12x + 7$ where $x \leq 2$

Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$$f^{-1}(x) = \dots\dots\dots$$

(Total for Question 25 is 4 marks)